DEFINITION *Centralization* in this chapter means physical and technological centralization—and never means logical centralization. Logically, every setup separates data products into their proper domains. But it is important to talk about physical and technological centralization because they sometimes make it harder to enforce or enable logical decentralization, which is what we want to achieve in the end.

Let's dive into the first architecture.

8.1 Data mesh on Google Cloud Platform

The self-serve data platform presented in this section is inspired by real-world examples showcased by Delivery Hero and Thoughtworks. Thoughtworks held workshops at Woman Who Code in 2020 and 2021 and presented an introduction to the data mesh using GCP, whereas Delivery Hero has presented a full-fledged enterprise-level data mesh.

What we're presenting here is mostly a middle ground. GCP is a great data platform to start off utilizing a lot of well-integrated services from Google Cloud. This will enable teams to quickly get off the ground. We address how to enlarge the data platform in this section as well as adding elements that are used at larger enterprise-level implementations.

In particular, the platform here puts a focus on having a good balance between being easy to set up and providing a certain level of autonomy to data-producing teams (unlike the Databricks-based platform you will see later).

8.1.1 Self-serve data platform architecture

Our data mesh has a few participants:

- *Black team*—A development team also tasked with creating and maintaining data products.
- *White team*—A data science team that creates and consumes data products.